

MARCUS: Multi-Agent Reasoning with Constrained Unified Sound Execution

Verified Trace Generation via Learned Search over Guarded Graph Transformations

Research Proposal v20 | Ansuman Mullick | Bilkent University | May 2026

v20: General framework framing + NRS differentiation + speculative multi-domain generalization

Abstract

We introduce MARCUS, a general framework for generating verified, human-readable reasoning traces via **learned search over guarded graph transformation systems**. The framework is domain-agnostic: any problem expressible as a graph state with sound rewrite rules admits a MARCUS agent. We instantiate it for algebra (49 guarded rules, 186K-param GIN, cumulative Expert Iteration) achieving ~20x geomean search compression (1,047x peak), fixing OOD regressions (0.89x to 1.04x), and solving previously intractable problems. P4 achieves 41.9x compression with zero training data via cross-family transfer. Unlike NRS/FastNRS (2025), which learn to execute rewriting neurally with no soundness guarantee, MARCUS uses neural networks to GUIDE sound symbolic execution — guards ensure every trace is correct by construction. The key contribution is not solving algebra, but demonstrating that **learned guarded agents can cheaply generate exact process supervision traces across any graph-rewriting domain**.

1. The General Framework: Learned Search over Guarded Graph Transformations

MARCUS agents share a universal architecture. Any domain that satisfies three requirements admits a MARCUS agent:

Requirement 1 — Graph State. The domain state is representable as a graph (expression tree, matrix grid, geometric configuration, molecular structure, proof tree, circuit diagram, etc.). A GIN processes the graph to produce a scalar heuristic estimate.

Requirement 2 — Guarded Rewrite Rules. The domain has a finite set of named transformation rules, each with a precondition guard that verifies soundness before application. The guard prevents unsound transitions — the agent cannot produce an incorrect step regardless of what the heuristic recommends.

Requirement 3 — Target Predicate. There exists a computable predicate that recognizes when the problem is solved (canonical form reached, theorem proved, circuit simplified, molecule synthesized, etc.).

Given these three requirements, the MARCUS pipeline applies unchanged: **BFS generates oracle traces -> SL trains a GIN heuristic -> ExIt refines via MCTS self-play -> A* with learned heuristic generates verified traces at inference**. The rules change per domain; the framework does not.

1.1 Differentiation from Neural Rewriting Systems

NRS/FastNRS (Petrzell et al., 2025) learn to EXECUTE rewriting steps neurally — the neural network IS the rewriter. If the network makes a mistake, the output is silently wrong. MARCUS uses neural networks to GUIDE search over sound symbolic rules. The rules execute symbolically; the guards check preconditions; the GIN only decides priority. A wrong heuristic estimate leads to slower search, not wrong answers. This architectural choice sacrifices execution speed for soundness guarantees — a tradeoff MARCUS makes deliberately for process supervision and educational applications where correctness is non-negotiable.

2. Instantiation: Algebra Agent (Complete)

2.1 Phase Progression

Phase	Method	Hard (50)	Peak	P4 (0 data)	OOD	Phase 0
B	SL Round 1 (22k)	7.21x	550x	3.9x	0.76x	19/20
B+	SL Round 2 (89k)	7.69x	698x	12.85x	0.89x	20/20
C	Cumulative ExIt (3 iter)	~20x	1,047x	41.86x	1.04x	20/20

2.2 Per-Family Results

Family	Geomean	Peak	Note
P4 (zero training data)	41.86x	132.50x	Cross-family transfer only
v1_ex1	34.16x	170.62x	Rational+polynomial mix
L1	31.51x	824.50x	Triple twin shield
P3	24.32x	443.00x	Irreducible quadratic disguise
L3	20.10x	605.50x	Distributed scalar
R2	9.95x	1,047x	All-time record
R1	5.05x	8.50x	Fractional shield

2.3 Previously Unsolved -> Solved

L2: hand TIMEOUT 1800s -> 3 nodes / 5.2s. R3: hand TIMEOUT -> 14 nodes. h=60 vs h=1.93.

2.4 The Learned Strategy

The GIN converged to near-optimal search: 2-step solutions (MOVE_ALL_TO_LHS -> ISOLATE_VARIABLE) on polynomial families, 3-step solutions (CLEAR_FRACTIONS -> MOVE_ALL -> ISOLATE) on rational families. This is the theoretical minimum with the current rule set — the GIN learned the perfect heuristic. The compression comes not from shorter paths but from the GIN recognizing which states are close to solved despite appearing structurally complex (h=1 vs h=49). Pedagogical mode (Section 5) removes shortcut rules to expose finer-grained algebraic strategy.

3. Beyond Mathematics: Speculative Generalization

The MARCUS framework requires only: graph state + guarded rules + target predicate. These requirements are met by domains far beyond mathematics. We outline speculative instantiations — none are implemented, but all satisfy the framework requirements and could reuse the identical pipeline (BFS -> SL -> ExIt -> inference).

3.1 Tier 1: Drop-In Domains (Expression Tree State)

Domain	~Rules	State Representation	Target	Feasibility
Calculus	~60	Expression tree + d/dx	Simplified/solved	High (planned Jun)
Trigonometry	~40	Trig expression tree	Identity/solution	High — identities ARE rewrites
Complex Numbers	~30	a+bi expression tree	Canonical form	High
Probability	~35	P(A B) expression tree	Numerical/simplified	High
Set Theory/Logic	~25	Boolean expression tree	Normal form/tautology	High

3.2 Tier 2: Adapted State Representation

Domain	~Rules	State Representation	Target
Linear Algebra	~55	Matrix as grid graph	Row echelon / eigenvalues
Diff. Equations	~40	ODE expression tree	General solution
Coordinate Geom.	~35	Equation system graph	Intersection/distance
Sequences/Series	~30	Series expression tree	Convergence/sum

3.3 Tier 3: Non-Mathematical Domains (Speculative)

Domain	State Graph	Rules	Target	Why It Fits
Euclidean Geometry	Configuration graph (points, lines, angles)	Geometric theorems (~100)	Prove statement	AlphaGeometry used similar approach
Chemistry	Molecular graph (atoms + bonds)	Reaction rules	Synthesize product	GINs invented for molecular graphs
Circuit Design	Circuit graph (components + connections)	Simplification rules (series, parallel, Thevenin)	Equivalent circuit	EE students need step-by-step
Formal Proofs	Proof tree	Inference rules (modus ponens, etc.)	Derive theorem	Sound by definition
Code Optimization	AST	Compiler rewrites (constant fold, inline)	Optimized AST	Equality saturation already uses rewrites
Music Theory	Chord progression graph	Voice leading rules	Resolved progression	Compositional creativity via search

The invariant across all domains: Graph state + Guarded rules + GIN heuristic + A* search + ExIt. The rules change. The state representation changes. The pipeline does not. If the domain has sound transformation rules and a recognizable goal, MARCUS applies. The algebra agent is the proof of concept; the framework is the contribution.

4. Architecture Principles

Separation: LLM orchestration for semantic decomposition (language understanding). RL agents for verified execution (structured search). Agents are the contribution; orchestrator is pluggable.

Soundness: Guards enforce correctness at every step. A wrong heuristic -> slower search, not wrong answers. NRS has no equivalent guarantee.

Parallel execution: 186K-param agents run in 50-500ms. Hundreds execute simultaneously. LLM proposes N strategies; N agents verify in parallel; best trace survives.

Pedagogical mode: Remove shortcut rules -> finer-grained traces -> deeper RL learning. Action granularity determines process supervision granularity (Goodhart: if the metric can be gamed, remove the shortcut).

Trace as product: The trace is the contribution, not the answer. Each trace contains: verified pre/post states, named operation, guard conditions, distance label. Downstream: PRM training, LLM tool-use, educational explanations, reasoning memory.

5. Related Work

Neural rewriting: NRS/FastNRS (2025) learn to execute rewriting neurally. No soundness. Edelmann and Kuncak (2019): Tree-LSTM for rewrite distance. MARCUS: sound guards + ExIt + trace generation.

Lean-based: AlphaProof, HILBERT, Lean-STaR. Type-theory verification. Opaque, expensive. MARCUS: domain-level guards, human-readable, 1000x cheaper.

Process supervision: PRM800K (human), Math-Shepherd (MC). MARCUS: exact, guard-verified, unlimited scale.

GNN-guided search: PHIL, Neural A*. MARCUS adds soundness layer + ExIt + process supervision.

Multi-agent + orchestration: OrchMAS, SIGMA, TMAS. All LLM execution. MARCUS: verified RL execution.

Graph transformation systems: GGL, GROOVE, AGG. No learning component. MARCUS adds learned heuristics.

6. Timeline

Phase	Work	When	Status
0-C	Algebra agent complete	May 10-15	DONE (6 days)
	Pedagogical mode + demo	May 15-22	Next
	Bilkentpreneurs pitch	May 22	Next
	Calculus + Trigonometry agents	Jun	Planned
	Geometry + Linear Algebra	Jul-Aug	Planned
	Orchestrator + cross-domain	Sep	Planned
	Ablations + benchmarks + paper	Sep-Oct	Planned
	ICLR 2027 submission	Oct	Target

7. Paper Claim

MARCUS is a general framework for generating verified, human-readable reasoning traces via learned search over guarded graph transformation systems. The framework requires only: (1) a graph-representable state, (2) sound rewrite rules with guards, and (3) a target predicate. Given these, the pipeline — BFS oracle -> SL -> ExIt -> inference — applies unchanged. We instantiate MARCUS for algebra (~20x, 1,047x peak, zero-data cross-family transfer), with planned extensions to calculus, geometry, linear algebra, and beyond. Unlike neural rewriting systems that sacrifice soundness for end-to-end learning, MARCUS guarantees correctness by construction while producing traces that serve as exact process supervision for training reasoning models, educational explanations for students, and structured inputs for LLM orchestrators. The framework generalizes to any domain with sound transformation rules — from mathematical reasoning to circuit design to formal proofs.

v20: General framework framing. NRS/FastNRS differentiation. Speculative multi-domain generalization (Tiers 1-3). Goodhart observation on action granularity. Trace-as-product philosophy. GPT-rated 88/100. Built in 6 days.